

Linna An 安琳娜, Ph.D in Chemistry

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EDUCATION

Ph.D. in Chemistry, University of Illinois at Urbana-Champaign (UIUC), U.S.A. **8/2014–12/2019**

Dissertation title: The biosynthesis and discovery of lanthipeptides

Committee: Wilfred A. van der Donk (Chair), Paul J. Hergenrother, Satish K. Nair, Douglas A. Mitchell)

B.S. in Chemistry, University of Science and Technology of China (USTC), China
09/2010–05/2014

RESEARCH EXPERIENCE

Tenure-track Assistant Professor , BioScience Dept, Rice University	01/2026–present
Affiliated Faculty , Bioengineering Dept, Rice University	04/2026–present
Affiliated Faculty , Chemistry Dept, Rice University	04/2026–present
Members of Ken Kennedy Institute , Rice University	03/2026–present
Members of Rice Synthetic Biology Institute , Rice University	04/2026–present
Adjunct Tenure-track Assistant Faculty , BioScience Dept, Rice University	08/2025–12/2025
Postdoctoral Scholar for Prof. David Baker , University of Washington	01/2020–12/2025
Graduate Research Assistant for Prof. Wilfred A. van der Donk , UIUC	08/2014–12/2019
Visiting Scholar with Prof. N. I. Martin , Utrecht University, the Netherlands	10/2015–12/2015
Research Assistant for Prof. Gaolin Liang , USTC	11/2011–06/2014
Team Member of team USTC_China, MIT iGEM 2011	10/2010–11/2011

PATENT

- D. Baker, **L. An**, S. Majumder, M. Said and L. Tran, 2023, DE NOVO DESIGNED SMALL MOLECULE BINDERS VIA EXTENSIVE SHAPE COMPLIMENTARY SAMPLING, 63/610,726.

PUBLICATIONS (#: equal contribution, *:correspondence)

- Design of Orthogonal Far-Red, Orange and Green Fluorophore-binding Proteins for Multiplex Imaging, L. Tran#, S. Klein#, S. Sharma, D. Jurgens, J. Decarreau, B. Liu, Y. Wang, A. Bera, A. Kang, J. Woods, E. Joyce, D. Vafeados, N. Roullier, W. Chen, G. Lee, L. Lavis, J. Mahamid*, **L. An***, D. Baker*. 2026, **Science**, doi: <https://doi.org/10.1101/2025.08.03.668343>. (Preprint link)
- W. Yang, S. Wang, G.R. Lee, J.Z. Zhang, A. Courbet, D. Juergens, X. Wang, T. Schlichthaerle, M. Abedi, R. Ragotte, L. An, I. Kalvet, S. Pellock, L. Mihaljevic, C. Glasscock, A. Pillai, A. Broerman, N. Ennist, E. Haefner, N. McNamara-Bordewick, I.

- Haydon, L. Stewart, G. Bhardwaj, & D. Baker, The past, present and future of de novo protein design, 2026 **Nature** 652 (8112) 1139–1152. ([Link](#))
3. I. Anishchenko, Y. Kipnis, I. Kalvet, G. Zhou, R. Krishna, S.J. Pellock, A. Lauko, G.R. Lee, **L. An**, J. Dauparas, F. DiMaio, & D. Baker, Modeling protein–small molecule conformational ensembles with PLACER, 2025 **Proc. Natl. Acad. Sci. U.S.A.** 122 (45) e2427161122. ([Link](#))
 4. G. Li[#], **L. An**[#], W. Yang[#], L. Yang, T. Wei, J. Shi, J. Wang, J. Doonan, K. Xie, A. R. Fernie, E. S. Lagudah, R. A. Wing, C. Gao^{*}. Integrated biotechnological and AI innovations for crop improvement, 2025, **Nature**, 643, 925. ([Link](#))
 5. J. Dauparas, G. Lee, R. Pecoraro, **L. An**, I. Anishchenko, C. Glasscock, D. Baker^{*}, Atomic context-conditioned protein sequence design using LigandMPNN, 2025, **Nature Methods**, Accepted, <https://doi.org/10.1038/s41592-025-02626-1>. ([Link](#))
 6. **L. An**^{*}, M. Seid, L. Tran, S. Majumder, I. Goreshnik, G. Lee, J. Dauparas, I. Anishchenko, B. Coventry, A. Bera, A. Kang, P. M. Levine, V. Alvarez, A. Pillai, C. Norn, D. Feldman, D. Zorine, D. R. Hicks, X. Li, M. G. Sanchez, D. K. Vafeados, P. J. Salveson, A. A. Vorobieva, and D. Baker^{*}. Binding and sensing diverse small molecules using shape complementary pseudocycles, 2024, **Science**, 385, 276. ([link](#)).
 7. **L. An**[#], D. Hicks[#], D. Zorine[#], J. Dauparas, B. Wicky, L. Milles, A. Courbet, A. Bera, H. Nguyen, A. Kang, L. Carter, D. Baker^{*}. Hallucination of closed repeat proteins containing central pockets, **Nat. Struct. Mol. Biol.**, 2023, 30, 1755. ([Link](#))
 8. **L. An**[#], D. Cogan[#], C. Navo, G. Jiménez-Osés, S. K. Nair, and W. A. van der Donk^{*}, Substrate-assisted enzymatic formation of lysinoalanine in duramycin, **Nat. Chem. Biol.** 2018, 10, 928. ([Link](#))
 9. **L. An**[#] and G. R. Lee[#], De novo protein design using the blueprint builder in Rosetta, *Current Protocols in Protein Science*, 2020, 102,1, e116. ([Link](#))
 10. **L. An**, and W. A. van der Donk^{*}, Recent progress in lanthipeptide biosynthesis, discovery, and engineering, *Comprehensive Natural Products III*. 2019, 2, 119. ([Link](#))
 11. **L. An**, M. Rehan H. Shah Gilani, G. L. Liang^{*}, Peptide-based nanostructures for cancer diagnosis and therapy, *Curr. Med. Chem.* 2014, 21, 2453. ([Link](#))
 12. J. Acedo, I. Bothwell, **L. An**, A. Truth, C. Frazier, W. A. van der Donk^{*}, O-Methyltransferase-mediated Incorporation of a β -Amino Acid in Lanthipeptides, *J. Am. Chem. Soc.* 2019, 141, 42, 16790. ([Link](#))
 13. **L. An**, “iGEM: Get & Give (& Share)”, *Microbiology Today*, Web. 15 May 2018. ([Link](#))
 14. Y. Liu[#], Q. Miao[#], P. Zou, L. Liu, X. Wang, **L. An**, X. Zhang, X. Qian, S. Luo, and G. Liang^{*}. Enzyme-controlled intracellular self-assembly of 18F nanoparticles for enhanced microPET imaging of tumor. *Theranostics*. 2015, 5, 1058. ([Link](#))
 15. Y. Yuan[#], L. Wang[#], W. Du, Z. Ding, J. Zhang, T. Han, **L. An**, H. Zhang, and G. Liang^{*}. Intracellular self-assembly of taxol nanoparticles for overcoming multi-drug resistance. *Angew. Chem. Int. Ed.* 2015, 54, 9700. ([Link](#))

16. Y. Yuan, J. Zhang, Q. Cao, **L. An**, G. Liang*. Intracellular disassembly of self-quenched nanoparticles turns NIR fluorescence on for sensing furin activity in cells and in tumors. *Anal. Chem.* 2015, 87, 6180. ([Link](#))
17. Y. Yuan, S. Ge, H. Sun, X. Dong, H. Zhao, **L. An**, J. Zhang, J. Wang, B. Hu, G. Liang*. Intracellular self-assembly and disassembly of ¹⁹F nanoparticles confer respective “Off” and “On” ¹⁹F NMR/MRI signals for legumain activity detection in zebrafish. *ACS Nano.* 2015, 9, 5117-5124. ([Link](#))
18. Y. Yuan#, H. Sun#, S. Ge#, M. Wang, H. Zhao, L. Wang, **L. An**, J. Zhang, H. Zhang, B. Hu, J. Wang, G. Liang*. Controlled intracellular self-assembly and disassembly of ¹⁹F nanoparticles for MR imaging of caspase 3/7 in zebrafish. *ACS Nano.* 2015, 9, 761. ([Link](#))
19. W. Wang#, J. Qian#, A. Tang#, **L. An**, K. Zhong, G. Liang*. Using magnetic resonance imaging to study enzymatic hydrogelation. *Anal. Chem.* 2014, 86, 5955-5961. ([Link](#))
20. Y. Yuan, J. Zhang, **L. An**, Q. Cao, Y. Deng, G. Liang*. Oligomeric nanoparticles functionalized with NIR-emitting CdTe/CdS QDs and folate for tumor-targeted imaging. *Biomaterials.* 2014, 35, 7881. ([Link](#))
21. Y. Yuan, S. Jiang, Q. Miao, J. Zhang, M. Wang, **L. An**, Q. Cao, Y. Guan, Q. Zhang, G. Liang*, Fluorescent switch for fast and selective detection of mercury (II) ions in vitro and in living cells and a simple device for its removal. *Talanta.* 2014, 125, 204. ([Link](#))
22. B. You*, P. Yin, and **L. An**. (2014), Multifunctional Electroactive Heteroatom-Doped Carbon Aerogels. *Small*, 10: 4352-4361. ([Link](#))
23. Y. Deng, Y. Luo, **L. An**, Y. Yue, M. Rehan H. Shah Gilani, G. Liang*. Covalently conjugating fluorescence probes to nanoparticles for signal enhancement, *Chem. Lett.* 2013, 42, 10, 130424. ([Link](#))
24. Y. Yuan., X. Wang, B. Mei, D. Zhang, **L. An**, X. He, J. Jiang, G. Liang*, Labeling thiols on proteins, living cells, and tissues with enhanced emission induced by FRET, *Sci. Rep.* 2013, 3, 3523 ([Link](#))

AWARD & FUNDING

1. CPRIT - Recruitment of First-Time, Tenure-Track Faculty Members (RFT) award, 2025/08 - 2029/08
2. Rice University BioSciences CARE Initiative Small Grants Program, 2026/01
3. Rice University Creative Ventures Fund: Conference and Workshop Development, 2026/04
4. RSBI 2026 Seed Grant, 2026/06

HONORS & PRESENTATIONS

5. 2024 Rising Stars in Biological Engineering, Princeton University, Omenn-Darling Bioengineering Institute, Princeton University, 09/2024
6. 2024 Rising Star in Engineering in Health, co-hosted by Boston University, Johns Hopkins

- University, Cornell University, Columbia University, 09/2024
7. 2024 UT Austin MBS Trailblazers of Tomorrow National Postdoctoral Symposium, 05/2024
 8. New Frontiers Scholar Award, Corteva, Inc. 08/2019
 9. Fuson Travel Award, University of Illinois at Urbana-Champaign, 07/2018

SELECTED Presentations

1. 2024 Workshop on Neutrons in Structural Biology, 06/2024
 - Invited talk: De Novo Small Molecule Binder and Sensor Design
2. RosettaCon, 08/2023
 - Presentation: De Novo Small Molecule Binder Design with Shape Complimentary Sampling
3. Biosynthesis, Biocatalysis, and New Methods in Enzymology, Gordon Research Seminar & Gordon Research Conference, 07/2023
 - Presentation & Poster: De Novo Small Molecule Binder Design with Shape Complimentary Sampling
4. MRSEC seminar, invited seminar. Brandeis University 06/2023
10. Rosetta Conference, Seattle (*Invited conference, supported by RosettaCon travel grant*) 08/2022
 - Poster: De novo design of small molecule binder using pseudocyclic proteins
11. Foresight's Molecular Machines Workshop, San Francisco (*Invited conference*) 07/2022
 - Invited Talk: De novo design of small molecule binding proteins
12. New Frontiers in Natural Products Discovery conference (*Supported by New Frontiers Scholar Award*), Corteva, Inc. 08/2019
 - Selected poster: A proof of concept target-guided lanthipeptide mining strategy
13. Natural Products and Bioactive Compounds Gordon Research Seminar & Gordon Research Conference (*Supported by Fuson Travel Award*), 07/2018
 - Talk & poster: Substrate-assisted enzymatic formation of lysinoalanine in duramycin
14. Midwest Enzyme Chemistry Conference, Loyola University, 10/2017
 - Selected talk: Substrate-assisted enzymatic formation of lysinoalanine in duramycin
11. Enzymes Coenzymes and Metabolic Pathways Gordon Research Seminar & Gordon Research Conference, 07/2017
 - Poster: Mechanistic details of duramycin biosynthesis
12. Midwest Enzyme Chemistry Conference, University of Illinois at Chicago, 10/2016
 - Poster: Mode of action studies for the two-peptide lantibiotic haloduracin
13. The 3rd National Chemistry Majors Scientific Activity Exchange (*Supported by the Department Travel Award*) Sichuan University, 12/2013

TEACHING & MENTORING

Mentor for undergraduate, Fiona Wong, Audrey Zhang, Rice, 01/2026-present

Mentor for high school student, Andrew Man, Rice, 01/2026-present
Mentor for postdoc, Wenbo Ning (Chemistry), Rice, 02/2026-present
Mentor for graduate students, Hayden Stagell (BCB), Yanapat Janathan (SSPB), Rice, 01/2026-present
Mentor for rotation students, Haotian(Odin) Zhang (computer science), UW, fall/2024-1/2025
Mentor for graduate students, Shajesh Sharma (bioengineering), UW, winter/2023-12/2025
Mentor for rotation students, Valentina Alvarez (biochemistry), UW, summer/2023
Mentor for graduate students, Long Tran (chemical engineering), UW, winter/2022-12/2025
Mentor for rotation student, Emma Mackey (biochemistry), UW, summer 2021
Mentor for graduate student, Chunyu Wu(biochemistry), UIUC, 09/2018-2019
Teaching Assistant for Enzymatic Reaction Mechanisms (graduate level) for Prof. J. A. Gerlt, UIUC, 01-05/2016
Teaching Assistant for General Chemistry (leading discussion and lab sessions) for Dr. K. Marville, and Dr. J. A. Martinez, UIUC, 01-05/2016, 01-06/2014, 08-12/2014
Graduate Mentor for Women Chemists Committee (WCC), UIUC, 2017

COMMITTEE

SERVICES

Gulf Coast Consortia Steering Committee, member, 04/2026-present
ML4BioChem Club, Co-founder and organizer, Houston, 01/2026-present
Women in Biology, Seattle chapter, 2022- present
Jupyter Summer Internship preparation committee, Institute for Protein Design, 2022
Women in Science, Institute for Protein Design, 2020- present
Conference Committee , Allerton Conference, UIUC, 2017
Women Chemists Committee, 2016 & 2015 Women Welcome Retreat, UIUC, 2016 & 2015
Women Chemists Committee, Stoesser Lecture, UIUC, 2016 & 2015

SKILLSETS

Computational skills: python (fluent), bash scripts, Fine tune of machine learning models, RosettaScripts (fluent), PyRosetta (fluent), machine learning-based protein design scripts (fluent), python-based data analysis (fluent).

Chemistry/Biochemistry related skills:

1. High throughput assay developments;
2. Peptide/Protein expression and purification using biological system or chemical methods;
3. Analytical methods development and use of HPLC, UV-vis, LC-HRMS (Q-TOF), GC-MS, MALDI, SEC;

4. Natural product discovery using bioinformatics & genome mining, semi-synthesis and purification, and structure elucidation using NMR and LC-MSⁿ;
5. Plasmid construction and gene manipulation for *Escherichia coli* and *Bacillus subtilis*.

Others:

1. Accomplished user of Illustrator, Photoshop, Autodesk 3D Max, Endnote, ChemDraw, html language;
2. Open water scuba diving;
3. Boulderling.